Answer Keys to Problem Set 2

Microeconometrics I with Joan Llull
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**This document is not intended to be a complete solution, but rather to provide some examples of interpretations and key points for the answer.*

Logit and Probit

- Estimate model

$$p(\boldsymbol{x}) \equiv \Pr[\text{Same industry} = 1|\boldsymbol{x}] = F(\boldsymbol{x}'\boldsymbol{\beta}),$$

	(1) Logit	(2) Probit
same_ind		
exper	-0.00233	-0.00148
	(0.00445)	(0.00277)
unempl_dur	-0.00278***	-0.00174***
	(0.000659)	(0.000410)
age	0.00767	0.00515
	(0.0340)	(0.0211)
age2	0.000508	0.000311
	(0.000557)	(0.000346)
female	0.133***	0.0822***
	(0.0340)	(0.0212)
ln_wage	0.262***	0.163***
	(0.0300)	(0.0186)
veneto_r~d	-0.199***	-0.124***
	(0.0374)	(0.0232)
_cons	-2.099***	-1.311***
	(0.537)	(0.334)
ll	-10662.5	-10662.6
N	15638	15638

Standard errors in parentheses
* p<0.05, ** p<0.01, *** p<0.001

Odds Ratios

```
Iteration 0:   log likelihood = -10818.448
Iteration 1:   log likelihood = -10662.625
Iteration 2:   log likelihood = -10662.549
Iteration 3:   log likelihood = -10662.549

Logistic regression                               Number of obs = 15,638
                                                  LR chi2(7)    = 311.80
                                                  Prob > chi2   = 0.0000
Log likelihood = -10662.549                       Pseudo R2     = 0.0144
```

same_ind	Odds ratio	Std. err.	z	P>|z|	[95% conf.	interval]
exper	.9976712	.0044377	-0.52	0.600	.9890113	1.006407
unempl_dur	.9972194	.0006571	-4.23	0.000	.9959325	.9985081
age	1.0077	.0342359	0.23	0.821	.9427838	1.077085
age2	1.000508	.0005574	0.91	0.362	.9994158	1.001601
female	1.141795	.0388167	3.90	0.000	1.068195	1.220466
ln_wage	1.299538	.039032	8.72	0.000	1.225245	1.378336
veneto_resid	.819524	.0306217	-5.33	0.000	.7616516	.8817936
_cons	.1225966	.0658511	-3.91	0.000	.0427826	.3513094

Note: _cons estimates baseline odds.

- Odds Ratio in Logit: e^{β} rather than β
- Example for interpretation: Increasing `exper` by one unit barely change (odds ratio $e^{\beta_j} \approx 0.998$, $p > |z| \approx$0.600) the decision to stay in the same industry (`same_industry` $= 1$). We can't distinguish its effect from zero ($e^{\beta_j} \to 1 \Rightarrow \beta_j \to 0$). One possible reason for this is that `exper` is general labor market experience, which is highly correlated with another regressor: age.

Marginal Effects

```
Conditional marginal effects                          Number of obs = 15,638
Model VCE: OIM

Expression: Pr(same_ind), predict()
dy/dx wrt:  exper unempl_dur age age2 female ln_wage veneto_resid
At: exper        = 11.50288 (mean)
    unempl_dur   = 15.69095 (mean)
    age          = 30.64529 (mean)
    age2         =  968.246 (mean)
    female       = .4156542 (mean)
    ln_wage      = 6.243519 (mean)
    veneto_resid = .7154368 (mean)
```

	dy/dx	Delta-method std. err.	z	P>|z|	[95% conf.	interval]
exper	-.0005812	.0011089	-0.52	0.600	-.0027547	.0015922
unempl_dur	-.0006942	.0001643	-4.23	0.000	-.0010161	-.0003722
age	.0019122	.0084699	0.23	0.821	-.0146885	.0185128
age2	.0001266	.0001389	0.91	0.362	-.0001457	.0003988
female	.0330579	.0084753	3.90	0.000	.0164466	.0496692
ln_wage	.0653194	.0074877	8.72	0.000	.0506437	.079995
veneto_resid	-.049619	.009315	-5.33	0.000	-.0678761	-.0313619

- The results from the mean marginal effects are consistent with the coefficient estimates. For example, for an individual with average characteristics (30 years old, 42% likely to be female, etc.), as experience increases by one unit, the individual is 0.05% less likely to remain in the same industry. This is barely an effect, we can't distinguish it from zero.

Multinomial Logit

```
Multinomial logistic regression                   Number of obs =  15,638
                                                  LR chi2(14)   = 2728.98
                                                  Prob > chi2   =  0.0000
Log likelihood = -14032.645                       Pseudo R2     =  0.0886
```

choice_ind	RRR	Std. err.	z	P>|z|	[95% conf. interval]	
Manufacturing	(base outcome)					
Services						
exper	.9413455	.0048604	-11.71	0.000	.9318673	.9509202
unempl_dur	1.007713	.0007979	9.70	0.000	1.00615	1.009278
age	1.203915	.0458359	4.87	0.000	1.117349	1.297189
age2	.9981284	.0006235	-3.00	0.003	.9969071	.9993511
female	1.650514	.0638165	12.96	0.000	1.530058	1.780453
ln_wage	.7479919	.025463	-8.53	0.000	.6997139	.7996009
veneto_resid	.9728996	.040549	-0.66	0.510	.8965844	1.055711
_cons	.120699	.0727368	-3.51	0.000	.0370462	.393245
Public_sector						
exper	.8622645	.0057761	-22.12	0.000	.8510175	.8736602
unempl_dur	1.019565	.0009598	20.58	0.000	1.017685	1.021448
age	1.743681	.1079043	8.98	0.000	1.544515	1.96853
age2	.9940603	.0009864	-6.00	0.000	.9921288	.9959956
female	6.626412	.3831103	32.71	0.000	5.916511	7.421492
ln_wage	.7437535	.0334216	-6.59	0.000	.6810502	.8122298
veneto_resid	1.021151	.0620432	0.34	0.730	.9065101	1.15029
_cons	.0000283	.0000279	-10.60	0.000	4.07e-06	.0001962

Note: **_cons** estimates baseline relative risk for each outcome.

- Example interpreting the relative risk ratio: an increase of one year of experience reduces the risk of working in the service sector after displacement by a factor of about 0.94 compared to the risk of working in manufacturing (base category), holding the other variables in the model constant.

Marginal Effects for MNL

```
Conditional marginal effects                          Number of obs = 15,638
Model VCE: OIM

Expression: Pr(choice_ind==Services), predict(outcome(2))
dy/dx wrt:  exper unempl_dur age age2 female ln_wage veneto_resid
At: exper        = 11.50288 (mean)
    unempl_dur   = 15.69095 (mean)
    age          = 30.64529 (mean)
    age2         =  968.246 (mean)
    female       = .4156542 (mean)
    ln_wage      = 6.243519 (mean)
    veneto_resid = .7154368 (mean)
```

	dy/dx	Delta-method std. err.	z	P>|z|	[95% conf.	interval]
exper	-.0083323	.0011017	-7.56	0.000	-.0104917	-.0061729
unempl_dur	.0010395	.000167	6.22	0.000	.0007121	.0013668
age	.0218789	.0083329	2.63	0.009	.0055469	.038211
age2	-.0002082	.0001364	-1.53	0.127	-.0004756	.0000591
female	.0447865	.0083423	5.37	0.000	.0284359	.061137
ln_wage	-.0552706	.0072666	-7.61	0.000	-.0695129	-.0410283
veneto_resid	-.0070239	.0090194	-0.78	0.436	-.0247017	.0106538

- You get one table for each sector.
- Examples of interpretation: For individuals with average characteristics, one additional year of experience decreases the probability of working in the service sector by 0.8%.

Conditional Logit

Wide-Form Data

- In the wide form data, you have a variable for experience in each sector, for example, `exp_manufacture` for manufacturing, which is not alternative-varying. For conditional logit, you need to transform the data from wide to long.

id	year	age	e_manuf_light	e_manuf_heavy	$\cdots$
1	1999	40	5	0	$\cdots$
2	1999	40	20	0	$\cdots$
3	1999	40	24	0	$\cdots$
4	1999	40	2	7	$\cdots$
5	1999	40	0	23	$\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$		

Conditional Logit

Long-Form Data

- In the long-form data, you have a variable, let's call it e_, for the sector-specific experience.

id	industry	d	e_	⋯
1	manuf_elecon	0	0	⋯
1	manuf_heavey	0	0	⋯
1	manuf_light	0	5	⋯
1	public_adm	0	0	⋯
1	public_hlth	0	0	⋯
1	serv_fin	0	0	⋯
1	serv_other	0	0	⋯
1	serv_sales	1	13	⋯
⋮	⋮	⋮	⋮	

- Now each row refers to a sector, you have a variable d indicating whether that individual chooses that sector or not, and the variable e_ indicates the individual's experience in that sector. The variable e_ is alternative-specific variable.

Conditional Logit

Collinearity

```
                                                  Wald chi2(43)   =   10720.73
Log likelihood = -21561.652                       Prob > chi2     =     0.0000
```

d_	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
industry						
e_	.2368502	.0027354	86.59	0.000	.2314889	.2422115
manuf_elecon	(base alternative)					
manuf_heavy						
unempl_dur	-.0148817	.0014249	-10.44	0.000	-.0176745	-.012089
age	-.0754779	.074813	-1.01	0.313	-.2221087	.0711529
age2	.0012089	.0012399	0.97	0.330	-.0012213	.003639
female	2.15553	.1623102	13.28	0.000	1.837408	2.473652
ln_wage	.0362148	.07177	0.50	0.614	-.104452	.1768815
veneto_resid	.5966138	.0767017	7.78	0.000	.4462812	.7469464
_cons	1.411464	1.175378	1.20	0.230	-.8922342	3.715163

- For conditional logit and nested logit, it is better to exclude general experience when including industry-specific experience in a regression.
- In fact, we don't include general cumulative experience not only here but also in the recent labor literature when we distinguish between job types.
- Some of the information in general experience is captured by industry-specific experience, and some is captured by age, education, and other related regressors.

Conditional Logit

Coefficients

Wald chi2(43) = 10720.73
Log likelihood = -21561.652 Prob > chi2 = 0.0000

d_	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
industry						
e_	.2368502	.0027354	86.59	0.000	.2314889	.2422115
manuf_elecon	(base alternative)					
manuf_heavy						
unempl_dur	-.0148817	.0014249	-10.44	0.000	-.0176745	-.012089
age	-.0754779	.074813	-1.01	0.313	-.2221087	.0711529
age2	.0012089	.0012399	0.97	0.330	-.0012213	.003639
female	2.15553	.1623102	13.28	0.000	1.837408	2.473652
ln_wage	.0362148	.07177	0.50	0.614	-.104452	.1768815
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_cons	1.411464	1.175378	1.20	0.230	-.8922342	3.715163

- Example of interpreting the coefficients: The estimate for the alternative-varying regressor e_ is positive ($\beta \approx 0.237$). This implies that an increase in experience in sector j leads to an increase in the probability of choosing sector j, while the probability of choosing the other alternatives decreases. We obtain consistent results with at-means marginal effects.

Conditional Logit

Marginal Effects

Pr(choice = manuf_heavy|1 selected) = .32037399

variable	dp/dx	Std. err.	z	P>|z|	[95% C.I.	]	X
e_							
manuf_elecon	-.003246	.000213	-15.22	0.000	-.003664	-.002828	.87006
manuf_heavy	.05157	.000756	68.18	0.000	.050088	.053053	2.6819
manuf_light	-.009298	.000267	-34.78	0.000	-.009822	-.008774	2.2968
public_adm	-.002716	.000162	-16.74	0.000	-.003034	-.002398	.31251
public_hlth	-.006653	.000235	-28.33	0.000	-.007113	-.006193	.67189
serv_fin	-.001704	.000113	-15.12	0.000	-.001924	-.001483	.18468
serv_other	-.020046	.000453	-44.24	0.000	-.020934	-.019157	1.0052
serv_sales	-.007908	.000247	-31.98	0.000	-.008393	-.007424	.83105

- We get consistent results from the average marginal effects: positive own effects, negative cross effects (and this is true for all sectors).
- Example of interpretation: A one-year increase in experience in heavy manufacturing is associated with a 5% increase in the probability of choosing that sector after displacement.

Conditional Logit

Subscripts of the Coefficients

- Recall the *additive random utitlity model* we have seen before,

$$U_j = X\beta_j + Z_j\gamma + \varepsilon_j, \;\; j \in \{1, \ldots, J\}, J > 2$$

- The response probability:

$$\begin{aligned} p_j(x, z) &\equiv \mathbb{P}(y = j | X = x, Z = z) \\ &= \mathbb{P}(U_j \geq U_k) \;\; , \forall k \neq j \\ &= \mathbb{P}(\varepsilon_k - \varepsilon_j \leq x(\beta_j - \beta_k) + (z_j - z_k)\gamma) \;\; , \forall k \neq j \end{aligned}$$

- For the individual-specific regressors X, why should I have β_j instead of β?
  - Think about this: Can β be identified if it's the same for all alternatives?
- Think about identification for the subscripts of all coefficients.
  - γ is always identified.
  - For βs, we have $J - 1$ differences to solve for J parameters, one of the errors need to be normalized. Only $J - 1$ errors of $\{\varepsilon_1, \ldots, \varepsilon_J\}$ are free to vary, and similarly, only $J - 1$ of $\{\beta_1, \ldots, \beta_J\}$ are free to vary.

Nested Logit

	(1) CLogit8	(2) NestedL~t
industry		
e_	0.237***	0.278***
	(0.00274)	(0.00477)
manuf_he~y		
unempl_dur	-0.0149***	-0.0226***
	(0.00142)	(0.00213)
age	-0.0755	0.00932
	(0.0748)	(0.0412)
age2	0.00121	-0.000516
	(0.00124)	(0.000716)
female	2.156***	3.454***
	(0.162)	(0.277)
ln_wage	0.0362	0.139
	(0.0718)	(0.0962)
veneto_r~d	0.597***	1.097***
	(0.0767)	(0.118)
_cons	1.411	
	(1.175)	

- Compared to the conditional logit, the coefficient estimates for the nested logit are slightly larger in magnitude. Part of the reason is that we don't have a bottom level intercept, the coefficients absorb it.

Nested Logit

Dissimilarity Parameter (McFadden (1978) and Maddala (1983))

/type		
manufact~u		1.597***
		(0.0502)
services~u		1.157***
		(0.0493)
public_tau		0.673***
		(0.0344)
ll	-21561.7	-21471.2
N	125104	125104

Standard errors in parentheses
* p<0.05, ** p<0.01, *** p<0.001

- For first-level choice j (one of the 3 sectors) and second-level choices k and l (industries), the dissimilarity parameter is

$$\sigma_j = 1 - \sqrt{1 - Cor(\varepsilon_{jk}, \varepsilon_{jl})}$$

The name dissimilarity parameter is intuitive as the parameter is a measure of the degree of independence in unobserved utility among the alternatives (industries) in the j-th limb (sector). $\rho_j = 1 - \sigma_j$ is called the *scale parameters* because they scale the regression parameters in the nested logit model.

Log Likelihood Function

Conditional Logit

Suppose we have m alternatives: $j = 1, \ldots, m$. The dependent variable y is defined to take value j if the j-th alternative is chosen:

$$y_{ij} = \mathbb{1}(y = j)$$

The probability that individual i chooses alternative j is

$$p_{ij} = \frac{e^{x'_{ij}\beta + z'_i\gamma_j}}{\sum_{l=1}^{m} e^{x'_{il}\beta + z'_i\gamma_l}}$$

We use the "mixed" formulation of the utilities for teaching purposes. The likelihood function is

$$\mathcal{L} = \sum_{i=1}^{N} \sum_{j=1}^{m} y_{ij} \ln p_{ij}$$

with all the components defined above.

Log Likelihood Function

Nested Logit

Suppose at the top level there are J limbs to choose from. The j-th limb has K_j branches. Denote by y_{ijk} the indicator for individual i choosing the branch k of limb j. The deterministic utility function is

$$\begin{aligned} V_{jk} &= \texttt{exper}'_{jk}\beta + w'\gamma_{jk} \\ &= x'_{jk}\beta + w'\gamma_{jk} \end{aligned}$$

The choice probability for branch k of of limb j is

$$\begin{aligned} p_j \times p_{k|j} &= \frac{\left(\sum_{s=1}^{K_j} e^{\frac{1}{\rho_j}\left(x'_{js}\beta + w'\gamma_{js}\right)}\right)^{\rho_j}}{\sum_{l=1}^{J}\left(\sum_{s=1}^{K_l} e^{\frac{1}{\rho_l}\left(x'_{ls}\beta + w'\gamma_{ls}\right)}\right)^{\rho_l}} \times \frac{e^{\frac{1}{\rho_j}\left(x'_{jk}\beta + w'\gamma_{jk}\right)}}{\sum_{s=1}^{K_j} e^{\frac{1}{\rho_j}\left(x'_{js}\beta + w'\gamma_{js}\right)}} \\ &\equiv \frac{e^{\rho_j I_j}}{\sum_{l=1}^{J} e^{\rho_l I_l}} \times \frac{e^{\frac{1}{\rho_j}\left(x'_{jk}\beta + w'\gamma_{jk}\right)}}{\sum_{s=1}^{K_j} e^{\frac{1}{\rho_j}\left(x'_{js}\beta + w'\gamma_{js}\right)}} \end{aligned}$$

$$\text{where} \quad I_j = \ln\left(\sum_{k=1}^{K_j} e^{\frac{1}{\rho_j}\left(x'_{jk}\beta + w'\gamma_{jk}\right)}\right)$$

Log Likelihood Function

Nested Logit

The FIML estimator maximizes

$$\mathcal{L}(\alpha, \{\beta_j\}_j, \{\rho_j\}_j) = \sum_{i=1}^{N} \sum_{j=1}^{J} y_{ij} \ln p_{ij} + \sum_{i=1}^{N} \sum_{j=1}^{J} \sum_{k=1}^{K_j} y_{ijk} \ln p_{ik|j}$$

In estimation, we might use sequential estimation (LIML) to provide starting values for FIML when its log-likelihood is not globally concave.

Nested Logit

Subscripts of the Coefficients

- Things are similar to that we have for the conditional logit.
- The deterministic part of the utility:

$$V_{jk} = X\beta_j + Z_j\gamma + Y_{jk}\alpha_j$$

- Why α_j? Consider this: if instead, we have α, can we identify ρ_j?

$$p_{jk} = p_j \times p_{k|j} = \underbrace{\frac{e^{X\beta_j + Z_j\gamma + \rho_j I_j}}{\sum_{m=1}^{J} e^{X\beta_m + Z_m\gamma + \rho_m I_m}}}_{\rightarrow\ \widehat{\beta_j}, \widehat{\gamma}, \widehat{\rho_j}} \times \underbrace{\frac{e^{Y_{jk}\alpha_j/\rho_j}}{\sum_{l=1}^{K_j} e^{Y_{jl}\alpha_j/\rho_j}}}_{\rightarrow\ \widehat{\alpha_j/\rho_j}}$$

where ρ_j is the dissimilarity parameter and I_j is the inclusive value or log-sum:

$$I_j = \ln\left(\sum_{k=1}^{K_j} e^{Y_{jk}\alpha_j/\rho_j}\right)$$

Nested Logit

Dissimilarity Parameter

- You can relate the dissimilarity parameter ρ_j to the *elasticity of substitution* you have seen in
  - utility functions (when there are at least two goods, CRRA is an example), and
  - production functions (when there are more than one inputs, capital-skill complementarity is an example).
- The grouping of alternatives is a modeling decision; alternatives with a high degree of substitutability should be placed in the same group.
- Therefore, even in Exercise 2, you don't have a regressor that affects the choice of the first level, the first level still exists.
- If a group has a single alternative, its dissimilarity parameter is not identified, so it should be set to one.
- Some authors interpret the nested model as a nested sequential choice, which may help structure the groupings but is not technically correct. The correct interpretation is degree of substituability, not the timing of decisions (textbook Hansen (2022), Ch26.6).